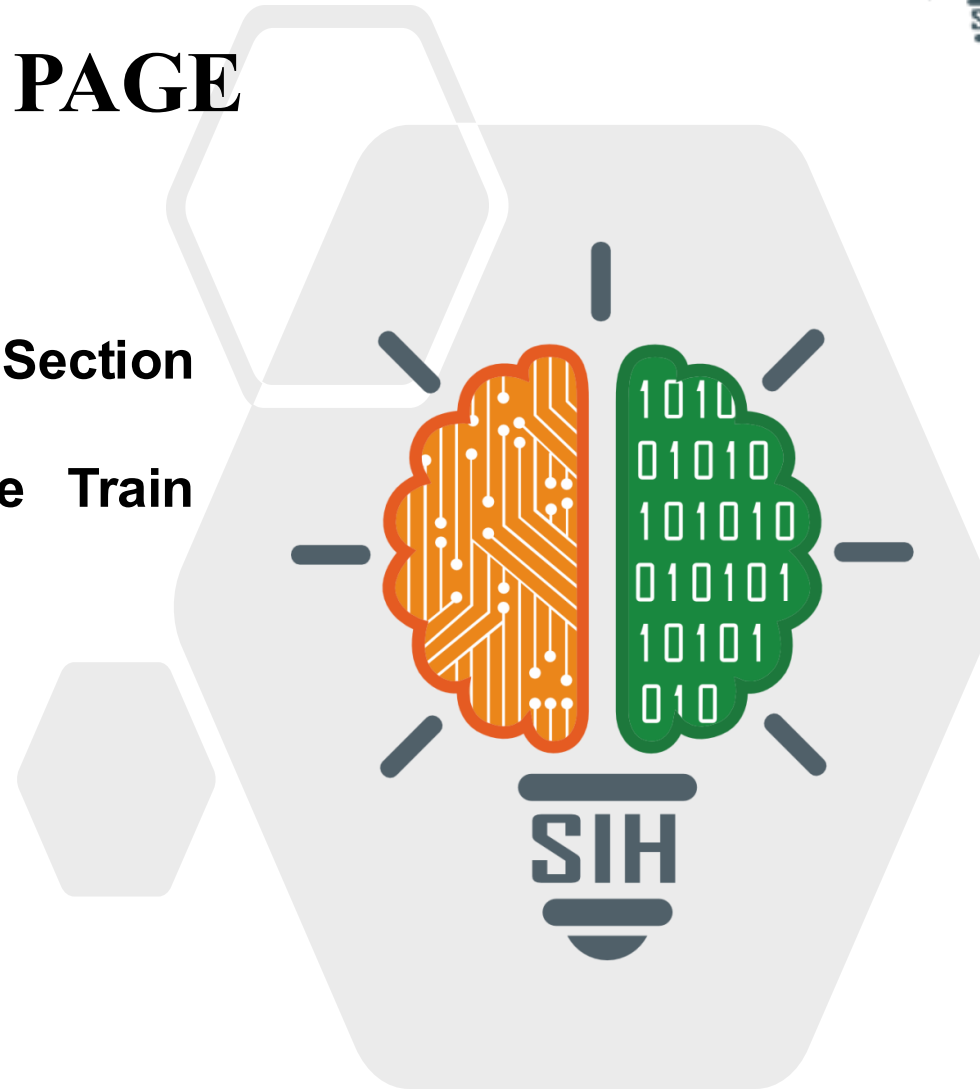


SMART INDIA HACKATHON 2025



TITLE PAGE

- **Problem Statement ID - 25022**
- **Problem Statement Title - Maximizing Section Throughput Using AI-Powered Precise Train Traffic Control.**
- **Theme - Transportation & Logistics**
- **PS Category- Software**
- **Team ID- 50575**
- **Team Name- The Avalanche**

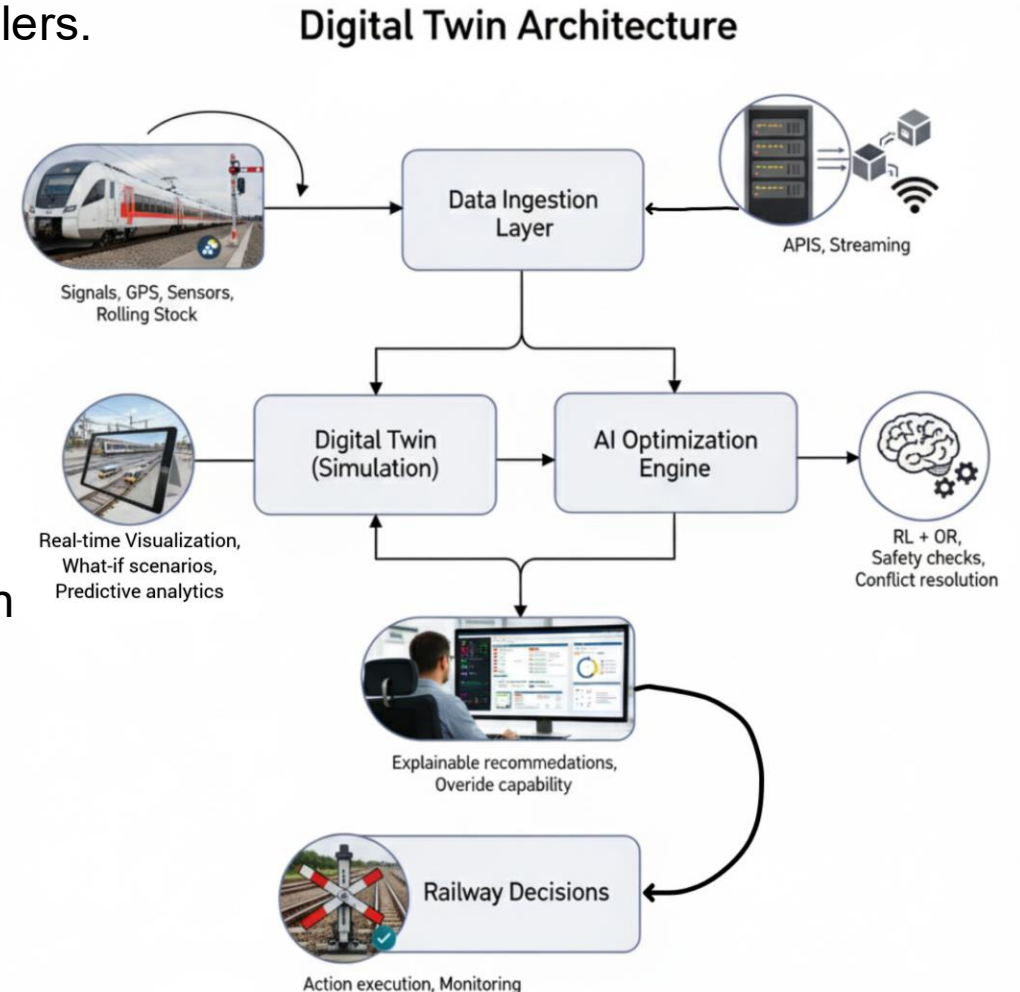


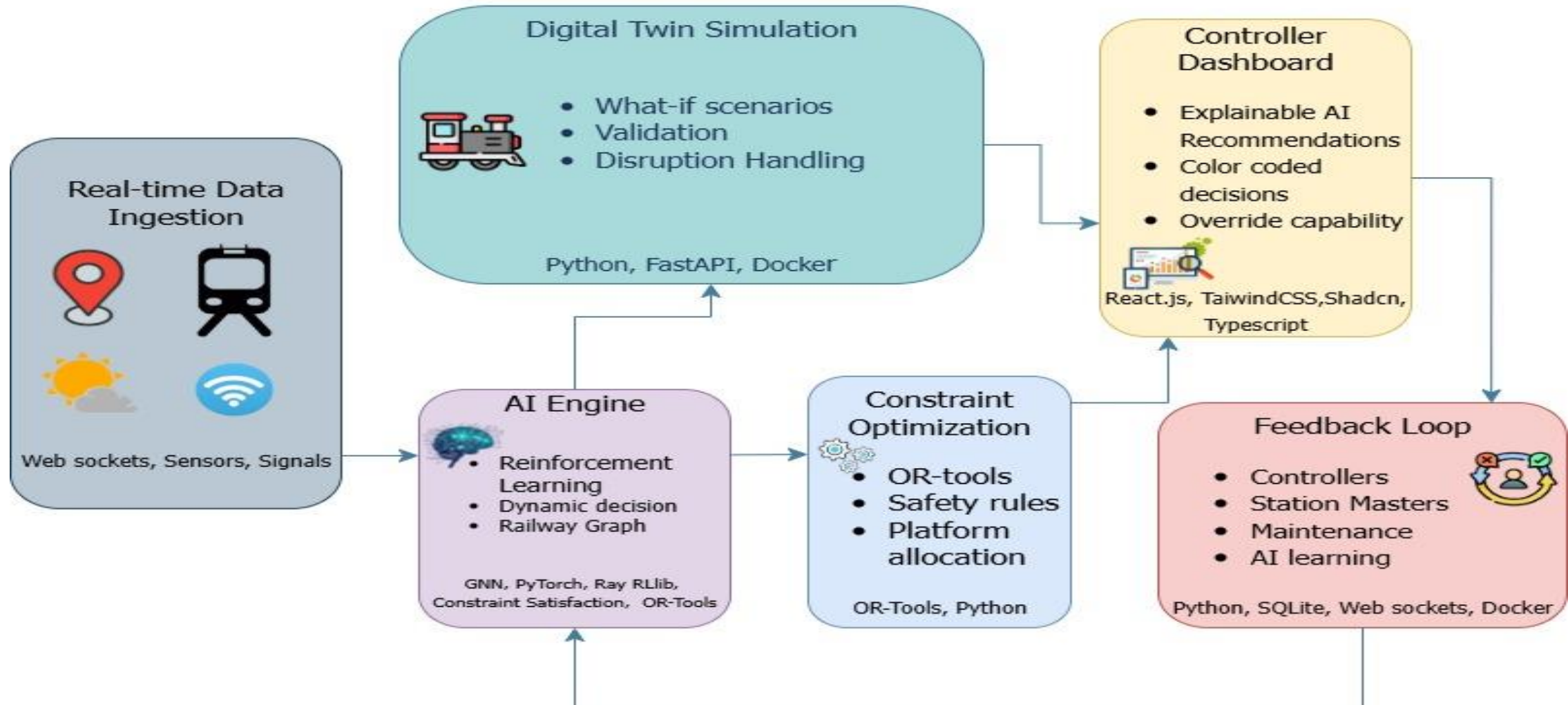
❑ RailAnukriti: AI-Powered Smart Train Traffic Optimizer

- Decision-Support System as a digital co-pilot for section controllers.
- Hybrid AI (RL + OR + GNNs) for real-time train precedence, crossings & platform allocation.
- Digital Twin for safe simulation & disruption handling.
- Explainable AI: recommendations with why & how, building controller trust.
- Adaptive learning from human overrides, delays, and past disruptions.

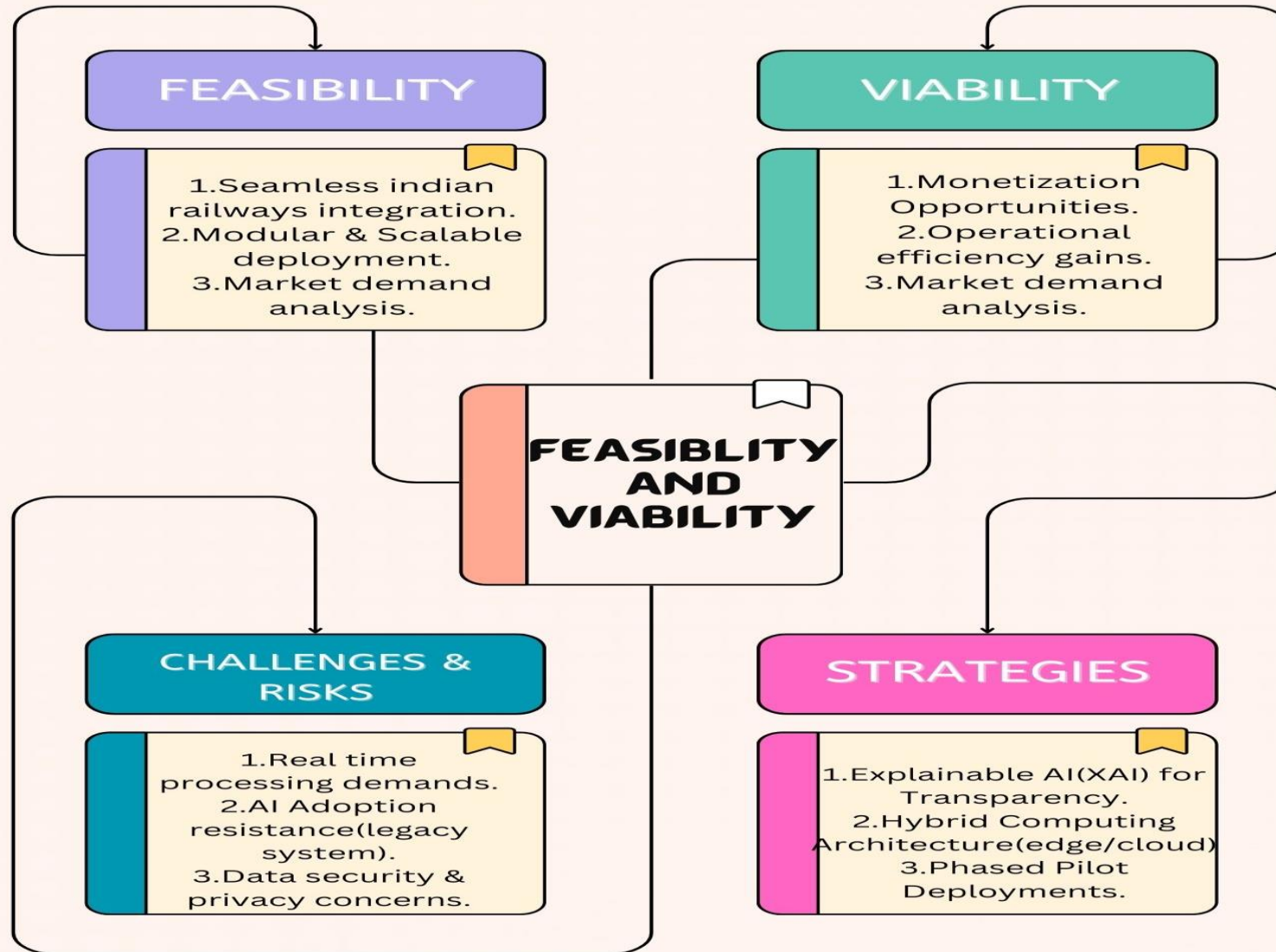
❑ Innovation & Uniqueness:

- Hybrid Intelligence: Deep RL + Constraint Programming + Graph Neural Networks.
- Generative AI-powered Scenario Assistant to simulate alternate plans.
- Human-in-the-loop: AI supports, controllers decide → ensures adoption & trust.
- Edge + Cloud AI for <1s responsiveness.

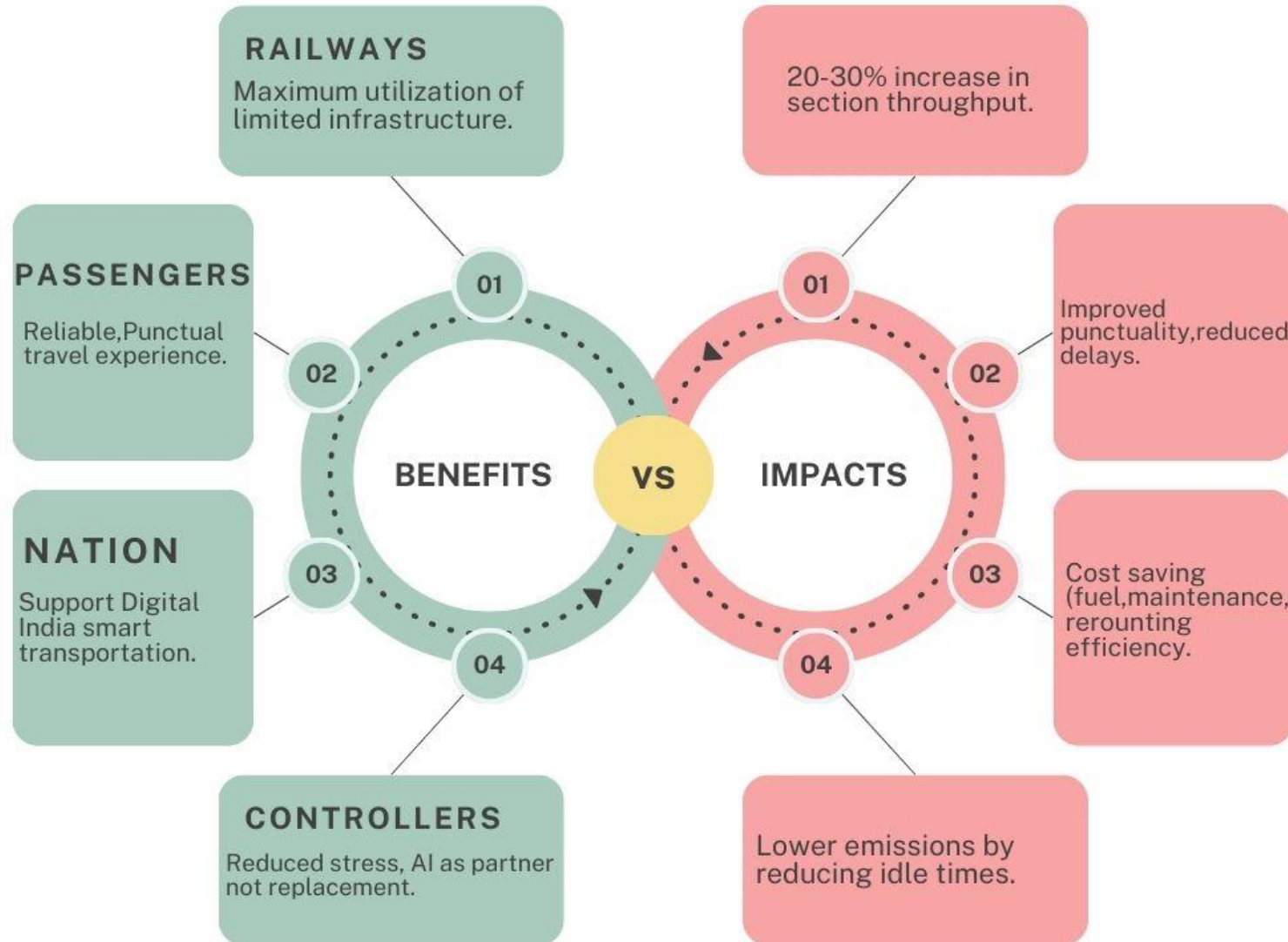




FEASIBILITY AND VIABILITY



IMPACT AND BENEFITS



- Google OR-Tools (Combinatorial Optimization).
- IEEE 2023 – AI for Train Scheduling under Uncertainty.
- Deutsche Bahn AI Rail Traffic Research.
- Indian Railways TMS & Signaling System docs.
- Reinforcement Learning + GNN studies (Elsevier, IEEE).

 [Youtube link](#)

 [GitHub Repository](#)

 [Live Link](#)

Research & References

